

- 4 Anika thinks that, for two square matrices $\mathbf{A}$ and $\mathbf{B}$, the inverse of $\mathbf{AB}$ is $\mathbf{A}^{-1}\mathbf{B}^{-1}$. Her attempted proof of this is as follows.

$$\begin{aligned}(\mathbf{AB})(\mathbf{A}^{-1}\mathbf{B}^{-1}) &= \mathbf{A}(\mathbf{BA}^{-1})\mathbf{B}^{-1} \\ &= \mathbf{A}(\mathbf{A}^{-1}\mathbf{B})\mathbf{B}^{-1} \\ &= (\mathbf{AA}^{-1})(\mathbf{BB}^{-1}) \\ &= \mathbf{I} \times \mathbf{I} \\ &= \mathbf{I}\end{aligned}$$

$$\text{Hence } (\mathbf{AB})^{-1} = \mathbf{A}^{-1}\mathbf{B}^{-1}$$

- (a) Explain the error in Anika's working. [2]
- (b) State the correct inverse of the matrix $\mathbf{AB}$ and amend Anika's working to prove this. [3]